

Spatially-Grounded Just-in-Time Episodic Memory for Mobile Robots

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Abstract—Dense semantic maps require minutes of per-scene pre-computation, yet a mobile robot needs an object’s location only when a task asks for it. We introduce *Just-in-Time (JIT) Episodic Memory*, which stores RGB-D keyframes behind a lightweight CLIP-FAISS index and defers detection, depth back-projection, and spatial clustering to query time. Our central result is a systems one: at a 1m operating point, deferred query-time geometry replaces a pre-built 3D semantic map with no loss of localization accuracy and orders of magnitude less pre-computation. JIT’s index builds in 3.1s against the 27–2,160s an eager map takes ($\sim 9\times$ cheaper than the closest antecedent, $\sim 700\times$ than the heaviest), a wall-clock build cost recovered after 18–860 queries per scene depending on the eager method, and later still on cumulative energy. On ScanNet v2 JIT reaches 81.3% Loc@1m (79.1% without its optional refinement stage), above every eager map at the same frame budget, and level with an equal-budget random-frame control on dense video, where the index’s role is the bounded detection budget rather than added accuracy. The CLIP-FAISS index bounds detection to a fixed k frames, holding query latency near-constant as the memory grows; its accuracy gain over random frame selection is regime-dependent, reaching +6 to +11 pp in a matched pipeline and largest on wide-baseline capture. One set of geometric hyperparameters carries four datasets (ScanNet, HM3D, ARKitScenes, and Replica), and the ranked candidates JIT returns place a correct location among the top three for 86.7% of queries, an output eager maps and single-prediction methods do not offer.

I. INTRODUCTION

A service robot that works in a home over days must answer spatial questions about it, such as “Where are my keys?” or “Is there a mug on the counter?” [1], [2]. Answering them calls for an *episodic memory* of what the robot has seen and where: a memory that is cheap to maintain on a power-limited platform, that returns an object’s 3D location rather than a picture of it, and that holds up as the home is rearranged from one day to the next. When that memory does its heavy perception, eagerly as the robot explores or lazily once a query arrives, largely decides whether it is practical to run on a robot at all.

Current systems sit at two extremes. *Eager* approaches (ConceptGraphs [3], VLMaps [4], LERF [5], LangSplat [6]) run heavy perception on every frame to precompute a 3D semantic representation: ready the moment a query arrives, but expensive to build, store, and keep current, and mostly wasted when few queries come. *Retrieval-only* methods, from visual place recognition [7], [8] to CLIP (Contrastive Language–Image Pre-training) embeddings over a FAISS (Facebook AI

Similarity Search) index [9], [10], are cheap but return the camera pose of a relevant view, not the object’s location. Neither extreme fits a robot that must stay cheap yet ground objects in 3D.

A middle path is newly practical because open-vocabulary foundation models have matured. A robot can store compact CLIP embeddings of its views online, then, only when a query arrives, run an open-vocabulary detector on the few views that a text query retrieves, so the expensive perception is deferred until a query aims it rather than run on every frame in advance.

We propose *JIT Episodic Memory* (Figure 1), which stores RGB-D keyframes and defers all geometric reasoning (detection, depth back-projection, clustering) to query time. CLIP-FAISS pre-filtering selects a small, relevant subset of frames so the expensive detect-and-project pipeline runs only where it is needed. Multi-view depth back-projection with DBSCAN (Density-Based Spatial Clustering of Applications with Noise) clustering then acts as a *spatial denoiser*: consistent cross-view observations converge on the true location while spurious detections scatter and are down-ranked.

Our contributions are:

- 1) A **deferred-computation architecture** for object localization that, at a 1m operating point, matches a strong per-query detection baseline and exceeds eager 3D maps built on the same frames, at orders of magnitude less pre-computation ($\sim 9\times$ to $\sim 700\times$, depending on the eager method) and with detection cost independent of memory size. One set of *geometric* hyperparameters serves 197 scenes across four datasets (Sec. V-D).
- 2) **Retrieval as a scalability mechanism**: CLIP-FAISS pre-filtering bounds detection to a fixed k frames, so query latency stays near-constant as the memory grows. Its effect on *accuracy* over random frame selection is regime-dependent, largest in wide-baseline capture but positive on dense video too in a matched pipeline; we significance-test and trace it across detector budgets (Sec. V-D).
- 3) **Ranked multi-hypothesis output**: query-time clustering yields spatially grounded candidates a planner can try in order, placing a correct location among the top three for 86.7% of queries (an oracle upper bound), +6.7 pp over a single-prediction random baseline that cannot rank at all, an output absent from eager maps and single-prediction baselines.

Spatially-Grounded Just-in-Time (JIT) Episodic Memory

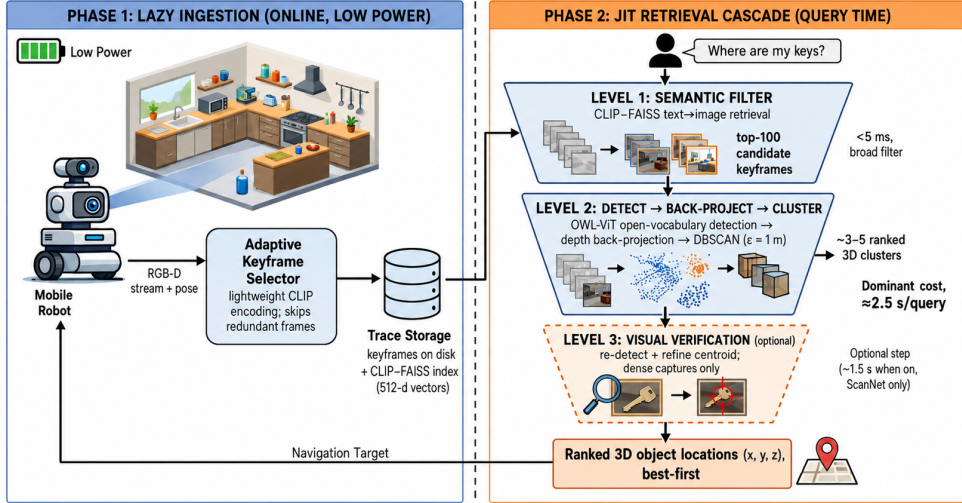


Fig. 1. JIT episodic-memory architecture. **Phase 1** lazily ingests RGB-D+pose into keyframes and a CLIP-FAISS index ($O(1)$ insert, $\sim 700\times$ less build-time pre-computation than an eager map). **Phase 2** answers a query with the three-level cascade shown (semantic filter; detect, back-project, cluster; optional verification), returning a ranked list of 3D locations.

II. RELATED WORK

a) Language-grounded 3D representations.: Eager methods precompute persistent 3D representations. ConceptFusion [11], VLMaps [4], and ConceptGraphs [3] fuse foundation-model features into maps or scene graphs; others distill vision-language features into dense 3D points [12], queryable maps [13], [14], radiance fields [5], [6], scene graphs [15], [16], Gaussian-splatting SLAM [17], or 3D instance segmentations [18], and some ground language via LLMs but need dense reconstructions as input [19], [20]. All trade upfront computation for fast lookup, requiring tens to thousands of seconds of build time and substantial per-scene storage (7–1,044 MB), and several require pre-built meshes unavailable in the episodic-memory setting. Even task-driven or online mapping systems [21]–[23] still maintain an incremental semantic map, whereas JIT retains only raw keyframes plus a CLIP index and materializes object geometry per query.

b) Episodic memory and object-goal navigation.: Tulving [24] introduced episodic memory; robotic episodic stores retain raw data for later retrieval. OpenEQA [25] reasons over episodic foundation-model memories to answer embodied questions, and GOAT [26] builds an object-instance memory during exploration and localizes goals by query-time matching. Retrieval-augmented robot memories such as ReMemBR [27] and Embodied-RAG [28] likewise store raw episodic observations and retrieve at query time, but reason in language to return text answers or waypoints rather than metric 3D object locations. The Ego4D episodic-memory benchmark [29] is the closest antecedent for *deferring* 3D localization to query time, and pinpoints JIT’s delta by how it splits the problem: its 3D task (VQ3D) localizes an *image*-specified instance within one

egocentric video, whereas its language task (NLQ) answers *text* queries with a temporal window rather than a 3D point. JIT keeps the store-then-localize-at-query principle but targets the combination Ego4D leaves uncovered—open-vocabulary text to metric 3D—trading image-crop queries and single-video structure-from-motion for text retrieval over a multi-scene metric RGB-D bank with ranked output. GOAT is the closest *robotic* antecedent, but it eagerly detects and commits object instances during collection, where JIT defers detection entirely. We benchmark GOAT head-to-head on all four datasets (Secs. V, V-D), where it is the strongest eager baseline on HM3D yet trails JIT at 1 m throughout. ObjectNav tasks the agent with reaching a named object [1], [30], with zero-shot CLIP methods [31]–[33], and OK-Robot [34] assumes a spatial grounding module; JIT provides the complementary *localization* from episodic memory.

c) Scene graph SLAM.: Hydra [35] and Khronos [36] build real-time 3D scene graphs but require continuous operation and dense sensor streams. JIT provides on-demand grounding from passively collected episodic data.

d) Active and anytime perception.: A long line of work treats *where the sensor looks* as the variable to optimize: active perception [37], [38], active vision [39], next-best-view planning [40], and anytime algorithms [41] that trade computation for quality on demand. JIT is complementary: it fixes what has been sensed and instead makes *when* perception is computed the variable, an orthogonal axis that could be combined with sensing-time control.

III. METHOD: JIT CASCADE

a) Problem formulation.: Given an episodic memory bank $M = \{(I_i, D_i, P_i, e_i)\}_{i=1}^N$ of N RGB-D keyframes (image I_i , depth D_i , pose P_i , CLIP embedding e_i) and a

Algorithm 1: JIT Cascade

Input: Query q , memory bank M , top- k , thresholds τ, ε
Output: Ranked list of 3D locations $\mathcal{L}$

```
// Level 1: Semantic retrieval
1  $e_q \leftarrow \text{CLIP}_{\text{text}}(q)$ ;
2  $\mathcal{F} \leftarrow \text{FAISS-top-}k(e_q, \{e_i\}_{i=1}^N)$ ;

// Level 2: Detect, project, cluster
3  $\mathcal{P} \leftarrow \emptyset$ ;
4 foreach frame  $f_j \in \mathcal{F}$  do
5    $\mathcal{D}_j \leftarrow \text{OWL-ViT}(I_j, q, \tau)$ ;
6   foreach detection  $d \in \mathcal{D}_j$  do
7      $z \leftarrow \text{DepthPercentile}_{30}(\mathcal{D}_j, d)$ ;
8      $\mathbf{p} \leftarrow \text{BackProject}(d, z, P_j)$ ;
9      $\mathcal{P} \leftarrow \mathcal{P} \cup \{(\mathbf{p}, \text{conf}(d))\}$ ;
10  $\mathcal{C} \leftarrow \text{DBSCAN}(\mathcal{P}, \varepsilon, \text{min\_samples}=1)$ ;
11 Rank clusters by  $S(c) = \sum_{(\mathbf{p}, s) \in c} s$ ;

// Level 3 (optional): Visual verification
12 foreach top-5 cluster  $c$  do
13   Re-detect query in representative frame (threshold  $\tau$ );
14   Refine centroid via detection-center depth re-projection
    (30th pctl);
15  $\mathcal{L} \leftarrow$  ranked cluster centroids;
16 return  $\mathcal{L}$ 
```

natural-language query q naming an object category, produce a *ranked list* of 3D locations $\mathcal{L} = [\mathbf{l}_1, \mathbf{l}_2, \dots]$ ordered by confidence, where $\mathbf{l}_1$ is the most likely object position. We assume known camera intrinsics and metric depth; poses may come from Simultaneous Localization and Mapping (SLAM) or a simulator.

b) Architecture.: JIT is a three-level cascade (Algorithm 1, Figure 1) that solves this problem by deferring detection and all downstream geometry (depth back-projection, clustering) to query time; we describe the three levels in turn.

Level 1 retrieves the top- k keyframes (default $k=100$) via CLIP-FAISS inner-product similarity [9], [10]. **Level 2** runs OWL-ViT [42] (Open-World Localization ViT; google/owlvit-base-patch32, $\tau=0.1$) on each retrieved frame, samples the 30th-percentile depth in a 30×30 patch around each detection centroid, back-projects to world coordinates, and clusters all projected points with DBSCAN [43] ($\varepsilon=1.0\text{m}$, $\text{min_samples}=1$). With $\text{min_samples}=1$, every point belongs to some cluster. Clusters are then ranked by cumulative detection confidence ($\sum s$): consistent multi-view projections merge into large, high-scoring clusters, while spurious detections form low-ranked singletons. This cumulative score also grows with cluster size, and we do not ablate alternative scorers. **Level 3 (refinement)** re-detects the query in cluster-representative frames and refines the centroids. Whether it helps depends on capture density: on dense video (ScanNet, $\sim 0.05\text{m}$ spacing) it sharpens fine-threshold localization and we enable it; on wide-baseline captures (HM3D, $\sim 3\text{m}$) it degrades accuracy and we disable it. A capture-density rule (spacing $< 0.5\text{m}$) reproduces this split automatically on both headline datasets, but it is not robust across regimes: L3 helps ARKitScenes

yet hurts Replica at *identical* selectivity (supplementary), so on the generalization datasets we default to the L1+L2 base and report L3 as an ablation (Table IV). A robust automatic selector across all regimes remains future work.

c) Online growth.: The FAISS flat index supports $O(1)$ insertion, so the bank grows incrementally with no rebuild, and a robot can answer queries immediately with whatever it has collected. Flat-index search is $O(Nd)$ but sub-millisecond at the sizes tested; the dominant cost is OWL-ViT detection on the fixed k frames, independent of N , so latency stays near-constant as the bank grows.

IV. EXPERIMENTAL SETUP

We evaluate on two complementary datasets with identical geometric hyperparameters.

a) Datasets.: **HM3D** [44]: 36 simulated validation scenes, 327 queries across 10 indoor categories (bed, cabinet, chair, couch, lamp, mirror, shelf, sink, table, toilet). We sample 125 random navigable positions $\times$ 4 views = 500 keyframes/scene in Habitat-Sim [45], [46] (simulator ground-truth poses, 90° horizontal field of view, HFOV), matching ScanNet’s frame count and giving JIT $k/N = 100/500 = 20\%$ selectivity on both datasets. **ScanNet v2** [47]: 141 real validation scenes, 457 queries, same 10 categories. We sub-sample 500 uniformly-strided keyframes from $\sim 2,500$ original frames (BundleFusion poses, 58° HFOV), giving JIT the same 20% retrieval selectivity. We evaluate 10 categories shared across all baselines for fair comparison; a 25-category expansion on HM3D (supplementary) confirms consistent trends with accuracy dropping for thin/flat objects. No *geometric* hyperparameters (DBSCAN, depth percentile, retrieval depth k) are re-tuned between datasets; the sole per-regime choice is the binary L3 refinement, on for dense ScanNet and off for wide-baseline HM3D (Sec. III). Cross-dataset generalization (Sec. V-D) uses the fixed L1+L2 base throughout, carrying no per-dataset choice.

b) Metric.: *Localization recall@ d* (Loc@ d): the top-1 prediction is correct if it lies within distance d of the *nearest* ground-truth centroid of the queried category. Being category-level, it is easier where a scene holds several instances, so absolute Loc@ d overstates instance-level difficulty: a uniformly random point already scores 42.6% at 1m on ScanNet against 4.4% on HM3D, and single-instance ScanNet queries sit near 69% for every lazy method (supplementary). Both effects are *method-invariant*, so comparisons are unaffected and only the absolute scale is inflated. Ground truth is matched by substring (ScanNet additionally expands synonyms); a stricter whitelist shifts all methods together (supplementary). All results are *per-scene macro-averaged* over 5 frame-sampling seeds; bootstrap confidence intervals and paired equivalence tests are in the supplement.

c) Baselines.: *Lazy methods*: **JIT**: the proposed cascade: CLIP retrieval $\rightarrow$ OWL-ViT on $k=100$ frames $\rightarrow$ depth-project $\rightarrow$ DBSCAN, with L3 refinement enabled per capture regime (on dense captures, median spacing $< 0.5\text{m}$; a manual setting, Sec. III). **Random-100**: OWL-ViT on 100 ran-

TABLE I

HM3D VAL (36 SCENES, 327 QUERIES, 500 KEYFRAMES/SCENE). JIT USES L1+L2 ONLY (L3 AUTO-DISABLED; SEC. III). AN EQUAL-BUDGET RANDOM-FRAME CONTROL REACHES 70.7% LOC@1 M HERE, 7.7 PP BELOW JIT (SEC. V). [†]DESIGNED FOR DENSE VIDEO; WIDE-BASELINE KEYFRAMES DEGRADE PERFORMANCE.

Method	Loc@0.25 m	Loc@0.5 m	Loc@1 m
<i>Lazy (map-free)</i>			
JIT	19.4%	56.2%	78.4%
<i>Eager (pre-built 3D map / instance memory)[†]</i>			
GOAT [26]	28.3%	44.9%	55.4%
VLMs [4]	3.2%	10.2%	19.6%
ConceptFusion [11]	17.1%	35.4%	48.8%
ConceptGraphs [3]	0.0%	1.0%	6.1%

dom frames $\rightarrow$ single best depth-projected detection. (Equal-budget random-selection control; we reserve *brute force* / BF-ALL for OWL-ViT on all N frames, Sec. V.) *Eager methods*: **ConceptGraphs** [3]: SAM ViT-H + CLIP ViT-H-14 $\rightarrow$ 3D object nodes (official codebase). **VLMs** [4]: per-pixel LSeg features [48] projected into a top-down grid with Gaussian radial weighting (official LSegEncNet + weighting formula). **ConceptFusion** [11]: SAM ViT-H masks + CLIP ViT-H-14 per-mask features with softmax-weighted pixel-aligned fusion, projected into a 3D point cloud (official formulation). **GOAT** [26]: the official Object Instance Memory (facebookresearch/home-robot), a Detic [49] detector committing object instances to a persistent memory by cross-view map association during collection; the category goal returns its most-observed instance and that instance’s 3D centroid, matching JIT’s single-prediction protocol. Localization only; GOAT’s navigation policy is out of scope. Hardware: RTX 4060 Ti 8 GB, i5-13600K, 32 GB RAM. Full baseline details, pose-only variants, and hyperparameter sweeps appear in the supplementary material.

V. RESULTS

We establish JIT’s accuracy against lazy and eager baselines on HM3D and ScanNet, then quantify its efficiency tradeoff (Sec. V), its generalization across four datasets and detector budgets (Sec. V-D), and its ablations.

A. HM3D (simulated)

Table I reports HM3D (36 scenes). JIT reaches **78.4%** Loc@1 m, **+7.7 pp** over an equal-budget random-frame control across all 5 seeds, its cross-view consensus preventing the single-view misprojections a single-frame baseline cannot recover from. Every eager baseline trails JIT by **23–72 pp**: wide-baseline keyframes (~ 3 m spacing) penalize dense-fusion maps that rely on frame overlap, whereas JIT’s per-query geometry does not.

B. ScanNet v2 (real-world)

At a 1 m operating point JIT leads every eager 3D map on ScanNet by 7–11 pp Loc@1 m while its index builds $\sim 9\times$ to $\sim 700\times$ faster (Table II; visualized in the supplement). The lead is specific to this operating point: at the stricter 0.5 m

TABLE II

SCANNet v2 (141 SCENES, 457 QUERIES, 500 KEYFRAMES/SCENE). BUILD AND STORAGE MEASURED ON OUR HARDWARE. THE **JIT** LOC@1 M IS L1+L2+L3, WITH THE PARENTHETICAL FOR L1+L2 ALONE (SEC. III). AN EQUAL-BUDGET RANDOM-FRAME CONTROL MATCHES JIT AT THIS OPERATING POINT AND IS ANALYZED AS A CONTROL IN SEC. V, NOT LISTED HERE AS A COMPETING MAP. STORAGE IS EACH METHOD’S OWN ARTIFACT; JIT’S 1.0 MB INDEX EXCLUDES THE ~ 375 MB OF KEYFRAMES IT RETAINS TO DEFER DETECTION (SEC. VI). [†]CG STORAGE IS THE SPARSE GRAPH. [‡]GOAT STORAGE IS ITS FULL INSTANCE MEMORY.

Method	Build (s) $\downarrow$	Storage (MB) $\downarrow$	Calls /q	Query (ms) $\downarrow$	Loc@1 m (%) $\uparrow$
<i>Lazy (map-free)</i>					
JIT	3.1	1.0 (+375)	100	2500	81.3 (79.1)
<i>Eager (pre-built 3D map / instance memory)</i>					
ConceptFusion	286	~ 250	0	25	74.1
VLMs	47	490	0	48	71.8
GOAT [26]	27	58 [‡]	0	<1	71.7
ConceptGraphs	2160	7 [†]	0	<10	70.3

threshold ConceptGraphs is sharper (Sec. V-D). Deferring geometry buys the 1 m result at the cost of query latency alone, ~ 2.5 s per query against the eager maps’ sub-50 ms. The equal-budget control confirms that lazy detection, not a pre-built map, is what carries the accuracy: OWL-ViT on 100 random frames (Random-100) reaches the same 80.7% (within 1σ of JIT) with no index at all. Both run the detector on exactly 100 frames, so JIT’s advantage is *which* frames CLIP selects, not how many, surfacing as accuracy rather than speed: it matches Random-100 on dense ScanNet video and pulls ahead on wide-baseline captures, reaching a given accuracy at up to $2.5\text{--}4\times$ fewer detector calls (Sec. V-D). Two eager methods show the failure modes JIT’s per-query geometry sidesteps. ConceptGraphs’ segment-and-merge pipeline fuses distinct objects as overlapping frames accumulate, and 10% of its queries return no prediction. GOAT, the closest antecedent to JIT and the strongest eager baseline on HM3D, reaches 71.7% here (9.6 pp below JIT) and builds fastest among eager methods (27 s), yet still pays roughly an order of magnitude more build time than JIT and must commit its perception to a persistent instance memory.

JIT’s ~ 2.5 s query is dominated by OWL-ViT; $k=50$ halves it for a ~ 3.5 pp Loc@1 m cost on HM3D, and faster detectors would cut it further at no accuracy cost (supplementary).

a) *Ranked candidate advantage.*: DBSCAN returns multiple ranked hypotheses, not a single point (Table III). JIT and Random-100 tie at the top-1, but the single-prediction baseline is *flat* in K while JIT’s ranked list climbs, so JIT leads by several points at $K=3$ and the advantage holds on single-instance queries where a one-point method structurally cannot rank (Recall@ K is an oracle upper bound on a planner that knows when it has succeeded). Against a *ranked* random+DBSCAN control the margin is smaller and single-instance-insignificant, so the defensible claim is a useful ranked list single-prediction methods cannot provide, not dominance over a clustered random baseline. Confidence order also beats proximity- and ratio-based selection (supplementary), and this output drives

TABLE III
RANKED LOCALIZATION ON SCANNET (RECALL@K AT 1 M,
PER-SCENE MACRO). FRACTION OF QUERIES WITH A CORRECT LOCATION
AMONG THE TOP-K RANKED CANDIDATES. RANDOM-100 SINGLE-BEST
RETURNS ONE POINT (FLAT IN K); RANDOM ROWS ARE 5-SEED MEANS
(STD ≤ 1.6 PP). *Single-instance* IS THE $n_{gt}=1$ SUBSET, WHERE THE METRIC
CANNOT REWARD HITTING THE NEAREST OF SEVERAL INSTANCES.

Method	All queries			Single-instance		
	R@1	R@3	R@5	R@1	R@3	R@5
JIT+L3	81.3	86.7	87.7	68.7	78.0	80.1
JIT (L1+L2)	79.1	83.9	84.2	69.7	76.9	77.3
Random-100 (single-best)	80.0	80.0	80.0	69.2	69.2	69.2
Random-100+DBSCAN	74.2	80.5	80.8	67.0	76.0	76.2

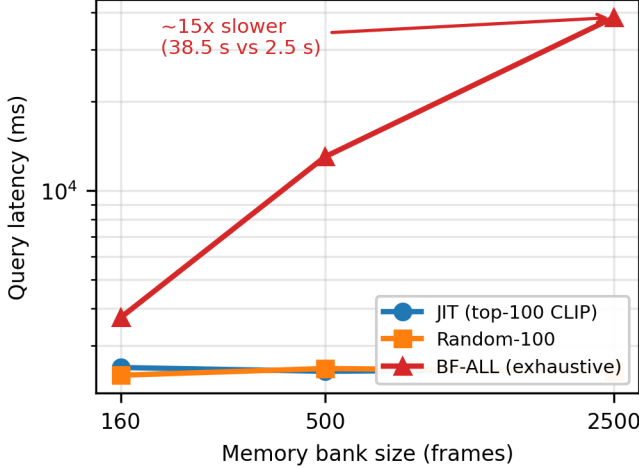


Fig. 2. **Query latency scales flat for JIT but linearly for exhaustive search.** Mean query latency versus memory bank size (both log; 160/500/2,500 frames) for JIT and Random-100 (fixed 100-frame working set) and BF-ALL (all N frames). JIT and Random-100 stay flat at ~ 2.5 s; BF-ALL grows to 38.5 s at 2,500 frames.

our live ObjectNav agent (Sec. V-D).

b) Scalability.: As the bank grows from 160 to 2,500 frames (20-scene subset, Figure 2), JIT holds $\sim 79\%$ Loc@1 m at *constant* 2.5 s query cost while exhaustive detection grows to 38.5 s ($\sim 15\times$): retrieval removes the linear per-query cost that memory growth would otherwise impose. A 141-scene selectivity sweep (supplementary) finds retrieval’s *accuracy* does not exceed equal-budget random selection at any bank size on this dense video, narrowing from a high-selectivity deficit toward parity as the bank grows: the index’s value here is the bounded budget, not accuracy.

C. Key ablations

Five ablations (supplementary) support the design. Exhaustive detection on every stored frame buys no accuracy over JIT’s retrieved 100 while costing $\sim 15\times$ the latency, so targeted retrieval already reaches that cost ceiling. DBSCAN adds +2.4 pp Loc@1 m over no clustering and yields the ranked list; the cascade is detector-modular but precision-limited (OWLv2 +8.4 pp, Grounding DINO -11.3 pp), with the Random-100 parity preserved under the stronger detector. L3 refinement

helps dense captures (Loc@0.5 m) and hurts sparse ones, hence the per-regime toggle, and JIT is robust to the DBSCAN, depth-percentile, and pose-noise choices (<1.2 pp each). A within-cluster centroid ablation (supplementary) confirms the plain mean is robust: confidence-capped means sharpen dense-capture Loc@0.5 m but regress elsewhere, so JIT keeps the mean.

D. Generalization and detector-budget efficiency

To probe whether JIT’s behavior holds beyond ScanNet/HM3D and to characterize *why* its lazy retrieval helps, we evaluate three additional axes under the same hyperparameters (Table IV): real consumer-LiDAR captures, synthetic captures, and detector budget. Retrieval plays two roles this section separates. Universally, it caps detection at a fixed k frames, holding query cost at $O(k)$ and latency flat as the bank grows. Its accuracy effect over random selection is regime-dependent, largest in wide-baseline capture.

a) Real consumer-LiDAR (ARKitScenes) and synthetic (Replica).: On both datasets JIT matches equal-budget random selection within seed noise (Table IV). These low-selectivity captures ($k=100$ covers most of the small bank) leave retrieval little room to help, so we report them as descriptive ties rather than certified equivalences (a formal test is underpowered; supplementary).

b) Detector-budget efficiency (HM3D).: Where the captures are wide-baseline, retrieval helps decisively. At realistic selectivity (HM3D, $k=100$ is 20% of the bank), CLIP-retrieved frames beat an equal-budget random-selection baseline by **+11.2 pp** Loc@1 m, positive on every seed (significance in the supplement). These equal-budget seeds vary the 100-frame selection within one fixed 500-frame bank; bank *composition* itself contributes ~ 3.9 pp (5 exploration seeds; supplementary) and a leave-one-scene-out check shifts the headline by <0.4 pp, so neither dominates the gain. A detector-budget sweep makes the mechanism explicit (Figure 3): JIT’s advantage is large across realistic budgets and closes only as k approaches full coverage, so JIT reaches random selection’s best accuracy at roughly $2.5\text{--}4\times$ fewer detector calls, and over-retrieving past $k=100$ does not help. High selectivity is necessary but not sufficient for a *large* gain. Holding the clustering pipeline fixed (JIT vs. Random-100, both DBSCAN-clustered, no L3), ScanNet’s like-for-like isolated retrieval gain is **+6.3 pp** macro Loc@1 m (79.1 vs. $72.8 \pm 1.1\%$, 5 seeds, well outside the random band), positive but below HM3D’s +11.2 pp, because dense real video makes adjacent frames more redundant. The larger HM3D gain thus reflects wide-baseline capture sparsity together with high selectivity; we read this as consistent with a capture-density mechanism rather than a controlled isolation of it (supplementary). This like-for-like gain differs from the headline ScanNet result, where the full system merely matches a single-best random control that forgoes clustering.

c) Lazy vs. eager grounding (ConceptGraphs).: Against the eager ConceptGraphs map on the same real-LiDAR captures, JIT matches CG at 1 m with no map build (differ-

TABLE IV

CROSS-DATASET LOC@1 M (%), PER-SCENE MACRO, ONE GEOMETRIC HYPERPARAMETER SET THROUGHOUT (THE JIT COLUMN IS THE FIXED L1+L2 BASE; JIT+L3 IS THE ADJACENT ABLATION). ARKITSCENES IS REAL CONSUMER-LIDAR (12 SCENES), REPLICA SYNTHETIC (8). RANDOM-100 AND RANDOM-100+DBSCAN (ALL DATASETS) ARE 5-SEED MEAN $\pm$ STD EQUAL-BUDGET CONTROLS. CG (CONCEPTGRAPHS) AND GOAT ARE EAGER BASELINES. SIGNIFICANCE TESTS IN THE SUPPLEMENT; HM3D/SCANNET REPEAT TABLES I AND II.

Dataset (scenes)	JIT	JIT+L3	Rand-100	Rand-100+DBSCAN	CG	GOAT
ARKitScenes (12, real LiDAR)	71.5	73.6	70.4 $\pm$ 1.7	71.7 $\pm$ 4.8	66.9	55.8
Replica (8, synthetic)	38.3	25.7	23.3 $\pm$ 2.8	35.4 $\pm$ 1.6	26.0	31.5
HM3D (36, sim)	78.4	—	70.7 $\pm$ 1.1	67.2 $\pm$ 2.0	6.1	55.4
ScanNet (141, real video)	79.1	81.3	80.7 $\pm$ 1.0	72.8 $\pm$ 1.1	70.3	71.7

Selectivity ($k=100$ of bank): ARKit 62%, Replica 62%, HM3D/ScanNet 20%. High selectivity is necessary but not sufficient: ScanNet’s shared 20% yields a +6.3 pp isolated-retrieval gain versus +11.2 pp on wide-baseline HM3D (Sec. V-D).

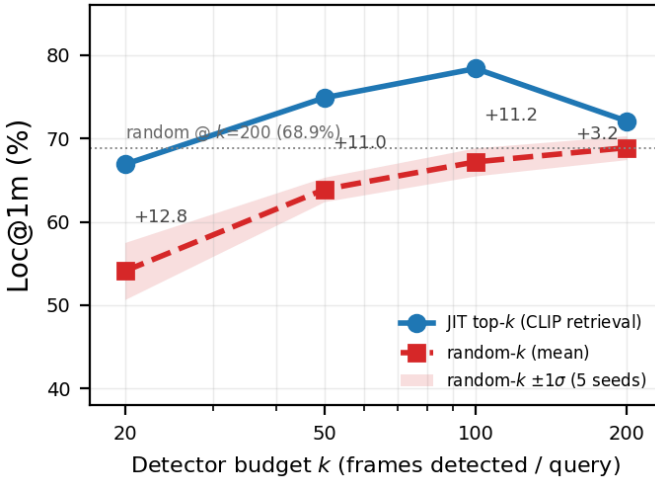


Fig. 3. Detector-budget efficiency on HM3D (Loc@1 m vs. per-query budget k , log scale). JIT top- k CLIP retrieval (blue) beats equal-budget random- k selection (red; 5-seed mean, $\pm 1\sigma$ band) at every budget (+11.2 pp at $k=100$, up to +12.8). JIT reaches random’s best (68.9% at $k=200$) by $k=50$, i.e. $\sim 4\times$ fewer detector calls.

ence not significant; supplementary), though CG is sharper at the strict 0.5 m threshold, as a pre-built map should be, at ~ 10 min/scene of build. On synthetic Replica the 1 m comparison favors JIT (not significant). GOAT, the closest antecedent, trails JIT at 1 m on all four datasets (Table IV).

d) Embodied feasibility (live ObjectNav).: To test whether ranked grounding transfers to a closed-loop agent, we run Habitat [45], [46] ObjectNav (327 episodes, 36 scenes): a random-walk explorer builds the bank online, queries JIT for a goal, and an oracle follower drives to it. JIT attains Success@2 m 38.5% (89.0% grounding); an oracle dense-coverage bank lifts this to 68.8%, the gap being an exploration tax plus a smaller prediction and follower residual (supplementary). The 38.5% is a *floor*: a learned explorer would raise the sparse random-walk coverage substantially.

e) Robustness to scene change (3RScan).: A memory kept over time must tolerate a *changing* environment. On 3RScan [50], which re-scans rearranged rooms, $\sim 45\%$ of la-

beled objects change between revisits, dominated by additions and removals (91 removed, 79 added, 23 moved) rather than sub-meter motion. An eager committed map cannot represent an addition or removal without rebuilding (~ 27 s for GOAT) on every revisit, whereas JIT refreshes by re-indexing (~ 3.1 s, $\sim 9\times$ cheaper), so deferred computation compounds its build-cost advantage over a robot’s lifetime; we do not claim a localization gain for the sub-meter moved objects.

VI. DISCUSSION

a) Deployment regimes: when to defer computation.

JIT and eager methods are complementary. An eager map pays a large one-time build and then answers in milliseconds, whereas JIT pays ~ 2.5 s per query but nothing upfront, so the wall-clock crossover falls at ~ 860 queries per scene against ConceptGraphs and sooner against lighter builds ($N \approx 18$ VLMaps, 114 ConceptFusion; supplementary); on cumulative *energy* it is far later, since JIT’s per-query detection is not amortized (supplementary), so its build-cost advantage does not extend to per-query energy. Eager maps are preferable in persistent, static deployments with frequent queries; JIT targets the complementary regime of environments that change between visits, infrequent or novel queries, and exploration, where a full map build would be immediately outdated.

b) Failure modes and detector dependency.: On HM3D the dominant failure at 1 m is *mislocalization* (21.4% of queries: the category is detected but its centroid lands beyond 1 m), while outright detection failure is rare (0.3%). Because the bottleneck is precision, not coverage, a stronger detector helps by producing tighter detections (OWLv2: +8.4 pp; supplementary). Instance-level analysis (supplementary) confirms most DBSCAN clusters are distinct physical objects rather than hallucinations.

c) Detector choice vs. deferred computation.: A natural concern is that JIT’s gain reflects its query-time OWL-ViT detector rather than the lazy architecture, since the eager baselines fuse only CLIP, SAM, or LSeg features. Refining ConceptGraphs’ own predictions post-hoc with OWL-ViT—a refine-only control over its committed centroids (118 of 457 queries; supplementary)—reaches only 69.7% Loc@1 m, **11.6 pp** below JIT. Part of this gap is fresh lazy detection on raw frames (Random-100 alone clears CG by ~ 10 pp); the remainder reflects the per-query frame selection and multi-view DBSCAN consensus a post-hoc detector on the committed map cannot recover.

d) Eager maps at native density.: All methods here share a 500-frame budget; one may ask whether eager maps close the 1 m gap given each scene’s full native density. Re-running ConceptGraphs at native ScanNet density raises its Loc@1 m from 70.3% to 90.7% on the scenes it can build, above JIT’s 79.1%. Building that map, however, takes hours and needs more than 20 GB of memory; JIT builds its index in 3.1 s and constructs no such map. Native density thus trades a heavy, often unaffordable build for accuracy JIT approaches at negligible cost. We therefore state the headline comparison

at the shared budget, where it is fair, and detail these runs in the supplement.

e) Evaluation scope.: Our category-level Loc@ d metric targets the episodic-memory setting (“find a chair” → ranked 3D hypotheses), distinct from referring-expression benchmarks (ScanRefer [51]); we compare only against methods that accept posed RGB-D keyframes and produce 3D locations, since HOV-SG [16] and OpenMask3D [18] require pre-built meshes unavailable here.

f) Deployment considerations.: Deployment on a physical robot raises three constraints we characterize in simulation, not on hardware: *compute* (~ 2.5 s/query, no on-robot measurement), *memory* (~ 375 MB of retained keyframes, larger than the eager artifacts but needing no build), and *failure recovery* (absent-object detection lets the robot refuse to navigate; supplementary).

VII. LIMITATIONS AND FUTURE WORK

Two scoping caveats bear on the headline claims. First, JIT’s ScanNet parity with Random-100 is refinement-dependent in sign (it edges the baseline with L3 and trails slightly without), so the honest reading is a statistical tie, on a category-level metric well above a random-in-scene floor (Sec. IV). Second, although CLIP retrieval improves a matched clustering pipeline on both datasets, the *full* system only ties a strong single-best random baseline on dense video, where retrieval’s guaranteed value is the fixed detection budget, not accuracy. Neither weakens the systems result, but both bound what the retrieval index itself buys. We evaluate 10 indoor categories; a 25-category HM3D expansion (supplementary) shows consistent trends, with thin and flat objects harder. Our primary ScanNet evaluation uses 500 frames/scene and the scalability analysis extends to 2,500, but a 10K+-frame evaluation would further strengthen the $O(k)$ claim. HM3D uses simulator poses; a pose-noise ablation, an SLAM-drift simulation (supplementary), and ScanNet’s real poses support viability, though deployment with a live SLAM system remains future work. The geometric hyperparameters were selected on a single scene but are robust (Loc@1m varies < 1.2 pp across every sweep, each value near its plateau center), and a leave-one-scene-out check confirms the headline is not a tuning artifact (supplementary); the one sensitive parameter is $k=100$, beyond which over-retrieving degrades accuracy. JIT assumes metric RGB-D depth (monocular estimators could extend it to RGB-only platforms with scale caveats) and uses CLIP ViT-B/32 (stronger encoders may help). For absent categories, thresholding the top-1 cluster score gives 0.90 AUROC present/absent classification (supplementary), a tunable operating point. JIT is presently category-level; compositional and relational queries (“the red chair by the window”) are a natural next step, resolved as a re-ranking pass over the top- K clusters using their retained frame provenance, which preserves the $O(k)$ query cost. A preliminary probe on Sr3D [52] supports this substrate: geometric re-ranking of JIT’s candidates lifts relational grounding from 42.4% to

54.0% (supplementary), the most direct route to referring-expression benchmarks [51]. Temporal fusion for dynamic scenes is a further extension.

VIII. CONCLUSION

We have shown that *when* geometric reasoning is performed is a design axis worth treating explicitly, alongside *how* it is performed. By deferring detection, projection, and clustering to query time, JIT Episodic Memory builds its index orders of magnitude faster than an eager 3D map (Table II), a cost recovered only after hundreds of queries per scene, and at a 1 m operating point it matches or exceeds every eager baseline while staying at parity with a strong equal-budget random-detection control, under one set of geometric hyperparameters across 197 real and simulated scenes. For exploratory mobile robots, where environments change between visits and queries are sparse, deferred computation is a natural fit: a robot answers spatial queries after seconds of CLIP encoding, without building a scene map, and its per-query cost falls as open-vocabulary detectors improve. Code is available at <https://github.com/kurshakuz/jit-memory>.

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